



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to JAYAVEL & SONS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

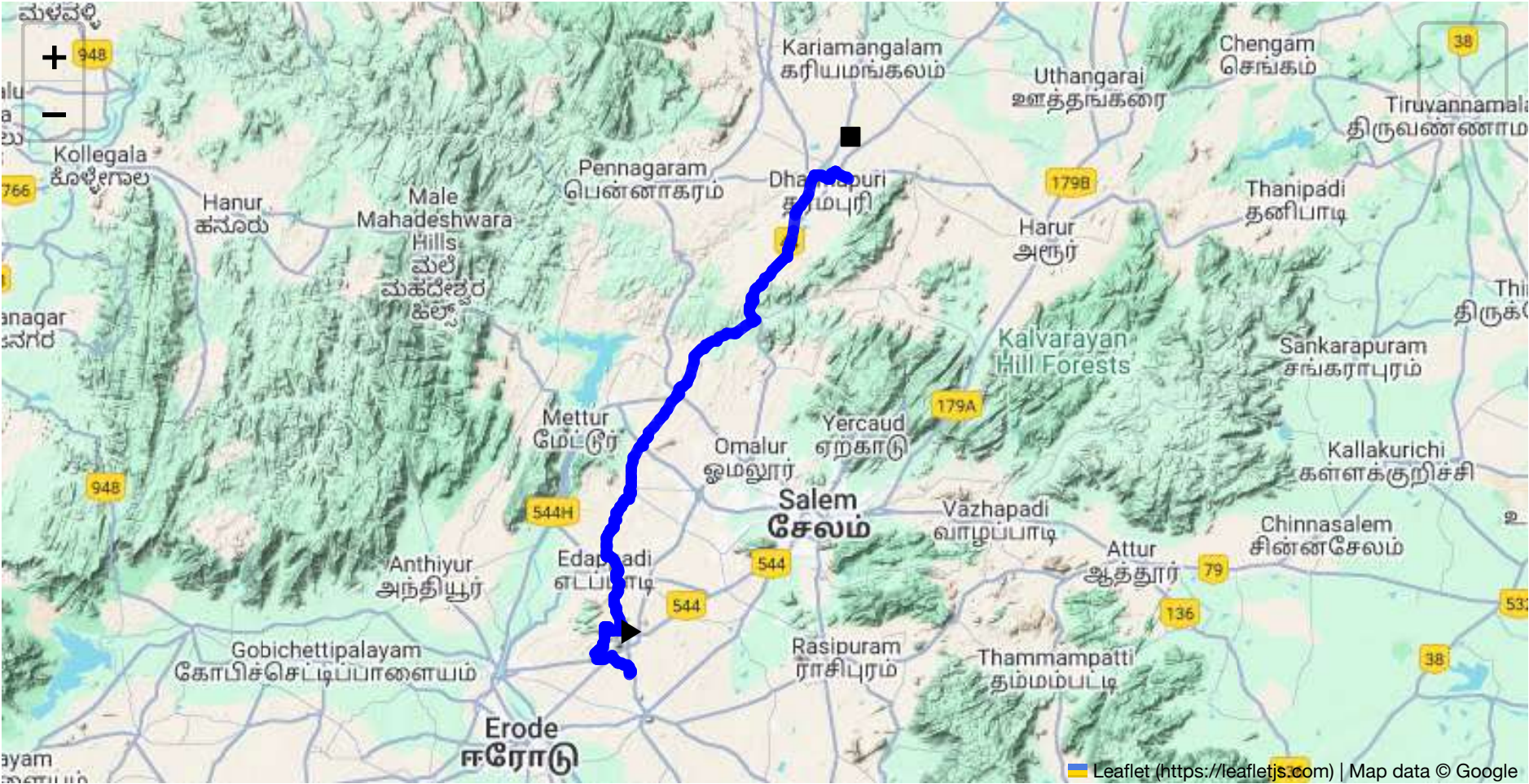
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 111.43 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours

Start: (11.4381, 77.8734)
End: (12.136872, 78.190429)



Welcome to the Journey Risk Management Study

Route Analysis Report

1. Overview of the Route Map

The route spans approximately 111.43 kilometers, starting from Sangagiri in Tamil Nadu and ending in Settikarai, with multiple waypoints including Chinnagoundanur, varuthampatti, Kidayur, Devannagoundanur, and Edappadi. The route primarily uses state highways and local roads, indicative of mixed urban and rural settings with potential variability in road infrastructure.

2. Typical Weather Conditions and Potential Weather-related Hazards

The region experiences tropical weather, with hot summers, significant monsoon rainfall from June to September, and mild winters. Monsoon season can cause flooding and road damage, potentially leading to hazardous driving conditions including poor visibility and slippery roads.

3. Traffic Patterns Analysis

Traffic congestion is common near urban centers such as Sangagiri and Edappadi, especially during morning (8-10 AM) and evening (5-7 PM) peak hours. Rural segments may have lighter traffic but can be prone to sudden slowdowns due to agricultural vehicles or livestock crossings.

4. Road Quality and Infrastructure

Road quality varies, with some well-paved sections and other stretches that may be narrower and less maintained, particularly on rural roads and intersections. Potholes and uneven surfaces are potential hazards in these areas, requiring vigilance.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or incidents, alternate routes might include:

- Utilizing major highways like NH44 for detours around congested areas.
- Using the Edappadi - Attur Road to bypass sections prone to heavy traffic. Drivers should ensure familiarity with alternate routes in low-traffic conditions.

6. Local Regulations Affecting Hazardous Material Transport

Local regulations require special permits for transporting hazardous materials. These are more strictly enforced in urban and sensitive areas. Compliance with signage, speed limits, and parking restrictions is crucial.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Incidents have been recorded near major intersections and busy corridors, often due to speeding or mechanical failures. Awareness and adherence to safety protocols are non-negotiable to mitigate risks.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural lands and potential wildlife zones, requiring careful management of potential spills or emissions. Drivers must adhere to environmental guidelines, especially when traversing zones indicated by signage.

9. Communication Coverage and Potential Dead Zones

Generally good communication coverage along urban routes; however, rural and hilly sections may experience weak signals. Pre-planned stops for communication checks are advised to ensure consistent contact.

10. Estimated Emergency Response Times

Urban centers like Sangagiri and Edappadi have quicker response times, estimated at 20-30 minutes. Rural segments may face delays, with response times extending to 45 minutes or more in remote areas. It's essential for drivers to have access to emergency contacts and a first aid kit.

11. Overall Summary of Risk Assessment

The primary risks include variable road quality, weather-related disruptions, and congestion near urban areas. Environmental and regulatory compliance is critical, particularly concerning hazardous materials.

In summary, while the route is manageable with proper precautions, drivers should remain attentive to changing conditions, have clear communication plans, and ensure all safety protocols are followed diligently. Assessment of real-time conditions, especially weather and traffic, prior to departure will enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Roundabout	High	12.13658, 78.16171	15 KM/Hr
3	Roundabout	High	12.14589, 78.17156	15 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	High	11.44900, 77.87406	15 KM/Hr
7	Turn	High	11.45353, 77.85697	15 KM/Hr
8	Turn	Medium	11.45781, 77.85203	30 KM/Hr
9	Turn	High	11.46307, 77.84941	15 KM/Hr
10	Turn	High	11.45733, 77.82861	15 KM/Hr
11	Turn	High	11.46528, 77.82475	15 KM/Hr
12	Turn	Medium	11.47230, 77.83209	30 KM/Hr
13	Turn	High	11.47232, 77.83296	15 KM/Hr
14	Turn	High	11.47551, 77.83497	15 KM/Hr
15	Turn	High	11.47601, 77.83427	15 KM/Hr
16	Turn	High	11.48030, 77.83486	15 KM/Hr
17	Turn	Medium	11.48074, 77.83599	30 KM/Hr
18	Turn	Medium	11.49399, 77.83833	30 KM/Hr
19	Turn	Medium	11.49482, 77.83896	30 KM/Hr
20	Turn	Medium	11.49973, 77.83871	30 KM/Hr
21	Turn	Medium	11.50037, 77.83816	30 KM/Hr
22	Turn	High	11.50415, 77.83805	15 KM/Hr
23	Turn	High	11.50438, 77.83843	15 KM/Hr
24	Turn	Medium	11.50090, 77.84541	30 KM/Hr
25	Turn	High	11.49948, 77.84585	15 KM/Hr
26	Blind Spot	Blind Spot	11.49869, 77.86563	10 KM/Hr
27	Turn	Medium	11.50748, 77.86206	30 KM/Hr
28	Turn	Medium	11.50866, 77.86053	30 KM/Hr
29	Turn	Medium	11.52587, 77.85524	30 KM/Hr
30	Turn	Medium	11.52722, 77.85576	30 KM/Hr
31	Turn	Medium	11.55039, 77.85511	30 KM/Hr
32	Turn	Medium	11.55481, 77.85639	30 KM/Hr
33	Turn	Medium	11.56366, 77.85965	30 KM/Hr
34	Turn	High	11.57903, 77.85441	15 KM/Hr
35	Turn	Medium	11.57910, 77.85555	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	Medium	11.58022, 77.85744	30 KM/Hr
37	Turn	Medium	11.60967, 77.84162	30 KM/Hr
38	Turn	Medium	11.61753, 77.84026	30 KM/Hr
39	Turn	Medium	11.62678, 77.84195	30 KM/Hr
40	Turn	Medium	11.63050, 77.84076	30 KM/Hr
41	Turn	Medium	11.64059, 77.84604	30 KM/Hr
42	Turn	Medium	11.64178, 77.84881	30 KM/Hr
43	Turn	Medium	11.64529, 77.85118	30 KM/Hr
44	Turn	Medium	11.65471, 77.85326	30 KM/Hr
45	Turn	Medium	11.66601, 77.85603	30 KM/Hr
46	Turn	Medium	11.67325, 77.86052	30 KM/Hr
47	Turn	Medium	11.67371, 77.86227	30 KM/Hr
48	Turn	High	11.68489, 77.86891	15 KM/Hr
49	Turn	Medium	11.68457, 77.87235	30 KM/Hr
50	Turn	Medium	11.68495, 77.87291	30 KM/Hr
51	Blind Spot	Blind Spot	11.69472, 77.87587	10 KM/Hr
52	Turn	High	11.69552, 77.87441	15 KM/Hr
53	Turn	High	11.76005, 77.89017	15 KM/Hr
54	Turn	Medium	11.76010, 77.89097	30 KM/Hr
55	Turn	High	11.75957, 77.89205	15 KM/Hr
56	Turn	Medium	11.83341, 77.94445	30 KM/Hr
57	Turn	Medium	11.83372, 77.94485	30 KM/Hr
58	Turn	Medium	11.84199, 77.94752	30 KM/Hr
59	Turn	Medium	11.90034, 77.98234	30 KM/Hr
60	Turn	Medium	11.91886, 78.01260	30 KM/Hr
61	Turn	Medium	11.91873, 78.02571	30 KM/Hr
62	Turn	High	11.93621, 78.05396	15 KM/Hr
63	Turn	High	11.93690, 78.05406	15 KM/Hr
64	Turn	Medium	11.95482, 78.04687	30 KM/Hr
65	Turn	High	11.95837, 78.04986	15 KM/Hr
66	Turn	High	11.96007, 78.04945	15 KM/Hr
67	Turn	Medium	11.96347, 78.05142	30 KM/Hr
68	Turn	High	11.96415, 78.05126	15 KM/Hr
69	Turn	Medium	11.96519, 78.04958	30 KM/Hr
70	Turn	High	11.96620, 78.04920	15 KM/Hr
71	Turn	Medium	11.96842, 78.05097	30 KM/Hr
72	Turn	High	12.14051, 78.13980	15 KM/Hr
73	Turn	Medium	12.13800, 78.15973	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
74	Turn	High	12.14485, 78.16498	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
1	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
2	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
3	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
4	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
5	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
7	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
8	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
9	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
10	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
11	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
12	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
13	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
14	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium
15	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
16	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
17	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
18	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
20	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
21	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
22	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
23	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
24	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
25	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
26	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
27	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
28	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
29	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.13658, 78.16171



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14589, 78.17156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



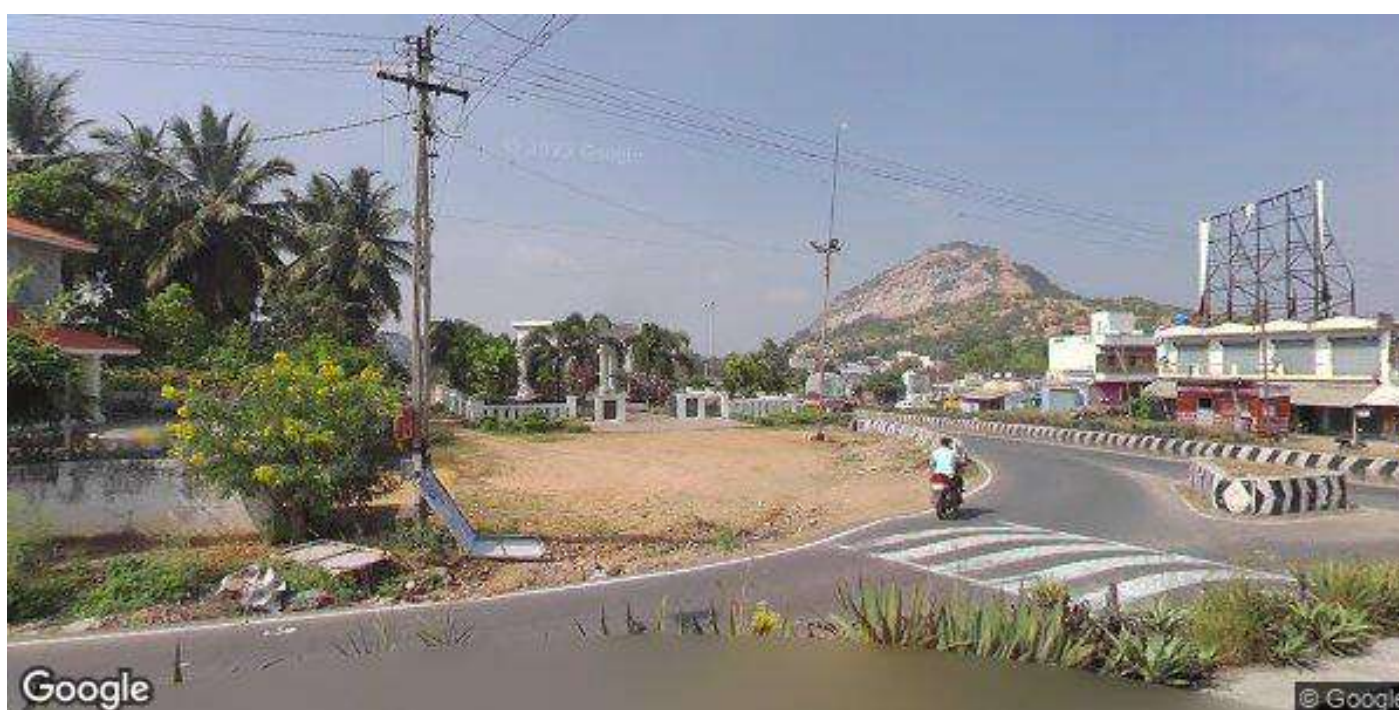
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45733, 77.82861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46528, 77.82475



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47230, 77.83209



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47232, 77.83296



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47551, 77.83497



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47601, 77.83427



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48030, 77.83486



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48074, 77.83599



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49399, 77.83833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49482, 77.83896



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49973, 77.83871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50037, 77.83816



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50415, 77.83805



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50438, 77.83843



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50090, 77.84541



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49948, 77.84585



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49869, 77.86563



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50748, 77.86206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52587, 77.85524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52722, 77.85576



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55481, 77.85639



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56366, 77.85965



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57910, 77.85555



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60967, 77.84162



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 77.84026



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62678, 77.84195



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63050, 77.84076



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64059, 77.84604



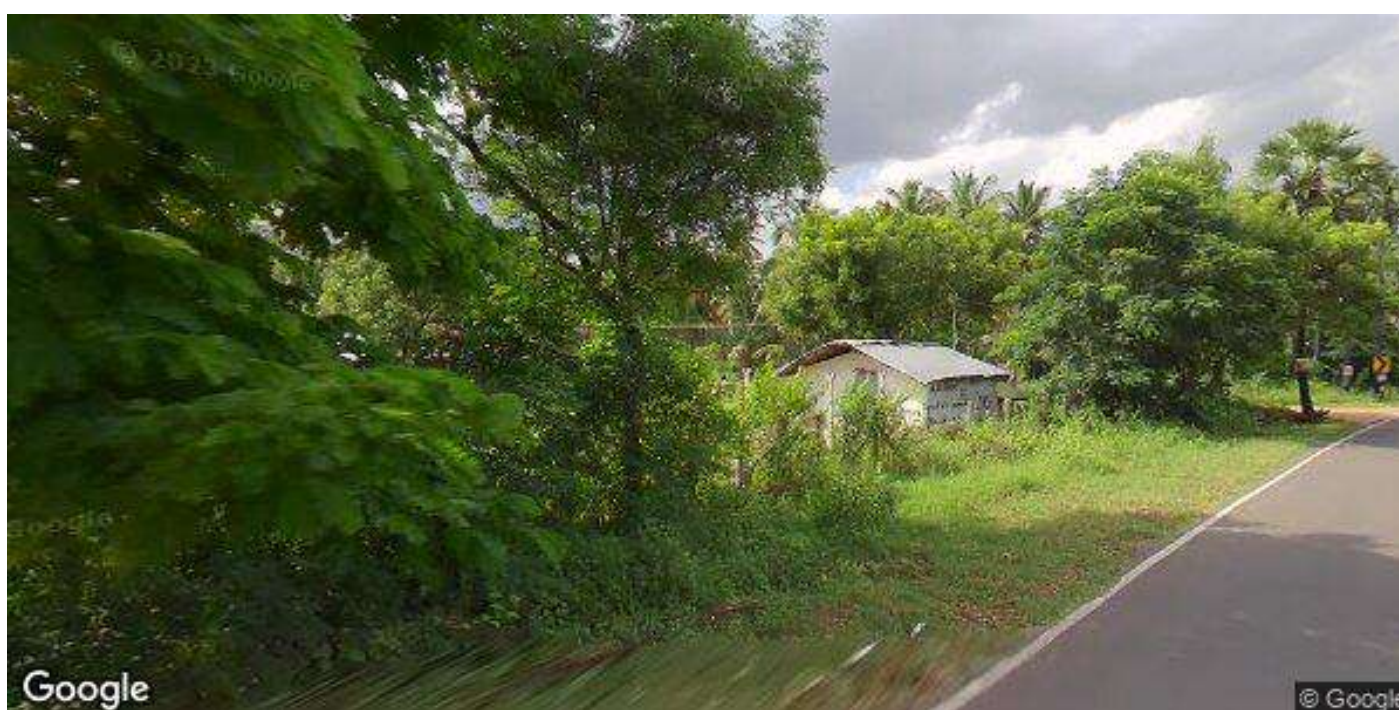
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64178, 77.84881



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64529, 77.85118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65471, 77.85326



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66601, 77.85603



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67325, 77.86052



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67371, 77.86227



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68489, 77.86891



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68457, 77.87235



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68495, 77.87291



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69472, 77.87587



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69552, 77.87441



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76005, 77.89017



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76010, 77.89097



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75957, 77.89205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83341, 77.94445



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83372, 77.94485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84199, 77.94752



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90034, 77.98234



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91886, 78.01260



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91873, 78.02571



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93621, 78.05396



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93690, 78.05406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95482, 78.04687



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95837, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96007, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96347, 78.05142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96415, 78.05126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96519, 78.04958



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96620, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96842, 78.05097



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14051, 78.13980



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13800, 78.15973



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14485, 78.16498

